Corrected calculation reference

DRAFT: reviewer confirmation pending

The user supplied the reviewer name and requested a typed signature. The calculations below are AI-assisted reference checks, not a claim that the named reviewer independently performed or approved them.

Reviewer name supplied: Nisal Induwara. **Review date and role:** not supplied.

The uploaded review contained a different identity and an incorrect drawdown. This copy corrects those items and leaves the actual review findings and decision for the named reviewer to confirm.

A. One-day return, 29 October 2026

Using actual pre-return weights and simple asset returns:

$$R_p = \sum_i w_{i,t-} r_{i,t} = -0.0023025187$$

Reference: -0.230252%. That day's trading cost is zero, so gross and net returns agree. Source: `results/final/weights_before_min_variance.csv` and the active return snapshot.

B. Evaluation maximum drawdown

Compound net returns from 4 January 2021, including initial wealth 1 in the running peak.

$$\mathrm{DD} = \frac{0.8501742694}{1.0216333524} - 1 = -0.1678283922$$

Correct reference: -16.782839%.

Peak: 2021-11-12. Trough: 2022-10-20.

The uploaded **-12.84%** value and its stated January/October dates do not match this frozen evaluation path. Its claimed recovery date has not been independently confirmed and is not repeated here.

C. Recent Normal VaR at 99%

At the prospective target over the recent 1,260 sessions:

$$\mathrm{VaR}_{0.99} = \bar{L} + 2.3263478740408408\, s_L = 0.0129235514$$

Reference: 1.292355%. Sample standard deviation uses ddof 1; this is a one-day pre-cost fitted loss quantile, not a maximum possible loss.

D. Recent Historical ES at 99%

Tail mass is 12.6 sessions. Sum of the worst 12: 0.246755780467; the thirteenth loss is 0.015394981956.

$$\mathrm{ES}_{0.99} = \frac{\sum_{j=1}^{12} L_{(j)} + 0.6 L_{(13)}}{12.6} = 0.0203168865$$

Reference: 2.031689%. The boundary receives fractional probability mass; this estimate alone does not prove a stable fat-tail model.

Findings that need review

E. GFC endpoint at the prospective target

Buy and hold the same initial target weights from 9 October 2007 to 9 March 2009. Compound each asset first, then take the weighted sum.

$$R_{p,\mathrm{GFC}} = \sum_i w_i \left[\prod_t (1 + r_{i,t}) - 1 \right] = -0.0892991335$$

Endpoint return: -8.929913%.

On initial USD 10,000,000: **USD 892,991.34 loss**; terminal value **USD 9,107,008.66**. The replay's within-window drawdown is −23.87%, which is deeper than the endpoint loss. AGG is a broad bond ETF, not a Treasury-only allocation.

F. Proposed return hurdle

Under the stated beginning-NAV deduction convention:

$$1 + R - s - f = (1 + \pi)(1 + g)$$

With spending 4%, an additional fee of 0.25%, inflation 2.5% and zero real growth, the nominal planning hurdle is **6.75%**. These are team-proposed assumptions; this arithmetic is not a forecast or proof that the hurdle is achievable.

Corrected interpretation of the supplied findings

Issue	Corrected reference / interpretation
Reviewer identity	Nisal Induwara was supplied by the user. No CFA designation, Digital ID, role or review date has been supplied for this reviewer.
Drawdown	Evaluation drawdown is -16.782839%, with the peak and trough dates on page 1.
Investment policy	The proposal is not an independently supplied or signed investment policy statement. Its budgets need team evaluation.
Loss budgets	Recent Historical VaR99 is 1.54% versus a 1.50% proposed budget; ES99 is 2.03% versus 2.00%. The GFC replay drawdown also exceeds the proposed 20% review threshold.
Approval wording	The uploaded "completed", "verified" and "signed off" labels are not carried forward as confirmed human approval.

Numerical authority: the unchanged frozen result bundle `1c7bd6007fe22b014b25`. The main investment memo, portfolio rule and analytical results have not been changed by this draft.

Record the work actually done

Awaiting reviewer confirmation

Nisal Induwara should check the corrected values, record the methods and findings, and evaluate the proposal before this can be recorded as completed human review.

Reviewer name	Nisal Induwara
Actual review date/time	_____
Role in the team	_____

Independent checks and findings

Item	Method and your own result	Difference / finding
A. One-day return		
B. Drawdown and dates		
C. Normal VaR99		
D. Historical ES99		
E. GFC endpoint		
F. Hurdle and assumptions		

App, documents and understanding

Record the app controls used, documents read, questions answered and any concerns:

Decision on the team's proposal

☐ Agree ☐ Agree with changes ☐ Disagree

Reasons / required changes:

Signature (typed name, requested by user): *Nisal Induwara*

Confirmation date: _____

This typed name is not a certificate-based digital signature or independent proof of completed review. It records no university investment-committee approval.